Term paper (blog post)

From weeks 8–14, you will be working on a short paper on a topic of your choosing. The papers are intended to be open-ended explorations related to any ideas from the course that you were curious to learn more about. You will be in a group of classmates writing on the same (or similar) topics, but you will write independently. You will post your writing (pseudonymously, if requested) on a class blog.

You will be guided through the blog post in each of your remaining homework assignments.

Timeline

Week 8. Pick group and (tentative) topic.

Week 9. Schedule 5–10 minute group meeting with professor to discuss topic. *(Six groups can schedule this before or after class or during the break.)*

Week 10. Submit outline and draft paragraphs.

Week 11. Submit progress report of draft and illustration.

Week 12. Submit draft blog post in markdown, and start planning final project video.

Week 13. Work on final blog post, and have ideas for final group project video.

Week 14. Submit final blog post, make final group project video.

Week 15. Present final group project video.

Guidelines

Your blog post must include an artifact that you generate, which can be a photo or video of something you made, a SVG you designed, a 3D model that you embed via Google model-viewer, or something in the same spirit.

You cannot use AI tools to generate any part of the text in your draft blog posts. On your final draft only, you can use a tool like Grammarly to check grammar, etc.

Throughout this project, you will talk to at least six people—which includes me and the two students in your group—about the project and ask them a question. The question can be about helping *you* understand an idea or it can be asking someone to tell you if *they* understand your explanation.

Your blog post should be between 1000 and 2000 words. (For reference, this document is roughly 1000 words.) Your target audience is your classmates at the start of week 8.

Write your blog post in markdown with MathJax (or in HTML). I will host it publicly on the class website. (You are permitted to use a pseudonym on request.)

You will upload your blog post to the website by following the instructions to do a pull request on Github.

In order to get credit, submit a PDF to Gradescope containing a screenshot of the pull request.

Possible topics

The following is a list of possible topics you could explore. You are encouraged to choose a topic that is *not* on this list.

Burnside's Lemma

Work in depth on a problem to count a family of objects using Burnside's Lemma, and perhaps add a new entry to the On-Line Encyclopedia of Integer Sequences.

Latin Squares

Cayley tables have the "Latin Square" property, which means that they behave a bit like Sudoku. Talk about Latin squares and their algebraic properties. Does every Latin Square correspond to a group?

Classification of Finite Simple Groups

In the 3Blue1Brown video we watched at the beginning of the term, Grant Sanderson talked about one of humanity's greatest intellectual accomplishments: the classification of finite simple groups. What are finite simple groups? Describe some of the groups in this family, such as the alternating group A_n or the finite simple groups of Lie type.

The Fundamental Group of a Topological Space

Groups show up in topology is a really cool way, where the elements of the group are certain kinds of paths on the space. Learn about these groups and their connection to the free group.

Polytopes as Cayley Graphs

We saw that Cayley graphs sometimes look like cubes (D_4), like hexagons (C_6), and even like a truncated octahedron (S_4). What other shapes arise from Cayley graphs?

Semidirect Products

One topic that we didn't explore in class is that of (outer) semidirect products, which allow you to write $D_n = C_n \rtimes C_2$. Describe how we can use these to combine two smaller groups into a new interesting group, such as the group of translations and 90° rotations of the plane, $\mathbb{Z}^2 \rtimes C_4$.

Lamplighter Group

Can we come up with a group $G = \langle S \mid R\rangle$ where S is the set of generators and R is the set of relations, but where S can be finite and R must be infinite? ("Finitely generated but not finitely related") The famous example of this is a cool application of *wreath products*.

Other Algebraic Structures

We've have (or will) study groups, abelian groups, rings, fields, and integral domains, but this term we won't talk about modules, vector spaces, algebras, semigroups, quasigroups, lattices, monoids, loops, and magmas. Tell us about one or more of these ideas.

Quaternion Group

Tell us more about the quaternion group, and maybe about how it generalizes into the octonions, and sedenions.

Frieze and Wallpaper Groups

There are 17 "wallpaper groups" and 7 "frieze groups" which are groups of repeating 2-dimensional patterns. Describe them, find as many examples of each as you can "in the wild" and show how they are groups.

Matrix Groups

Tell us about the general linear group, special linear group, and projective linear group and how they can be used in robotics and other applications.

Word Problem for Groups

Given a group presentation $G = \langle S \mid R\rangle$ and a word with letters in S, is that word the identity? This problem is *unsolvable* in general, meaning that there is no general purpose algorithm for solving this problem. What's going on with this?

Coxeter Groups

We can understand the symmetries of an object in space in terms of reflections or kaleidoscopes. Tell us about some of these. Perhaps focusing on the icosahedral group.

Sylow Theorems

Sure you believe me when I tell you that there are five groups of order 8 up to isomorphism, but how do you *prove* it? You use the Sylow theorems!

Elliptic Curves

You can define a group on Elliptic curves, which are used in Elliptic Curve Cryptography and in the proof of Fermat's Last Theorem. What does this group look like?

Exceptional Outer Automorphism of S_6

Describe the automorphism group $$ \operatorname{Aut}(G) = \{\varphi\colon G \to G \mid \varphi \text{ is an isomorphism}\} $$ and in particular, look at the exceptional outer automorphism of S_6.

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